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Cambridge IGCSE™ Combined and Co-ordinated Sciences
Second edition

C4 Electrochemistry

Pages 326–342

Comprehensive Exam Preparation Guide

C4 Electrochemistry — Comprehensive Study Guide

Cambridge IGCSE Combined and Co-ordinated Sciences (Second Edition) | Pages 326–342

1. Chapter Overview

This chapter covers **electrochemistry** — the relationship between electricity and chemical change. It is split into two main topics:

- **C4.01 Electrolysis** — Using electricity to decompose ionic compounds, the components of an electrolytic cell, the products formed at the electrodes, and how to write ionic half-equations.
- **C4.02 Hydrogen–oxygen fuel cells** — Converting chemical energy into electrical energy, how fuel cells work, and their advantages and disadvantages compared to petrol engines.

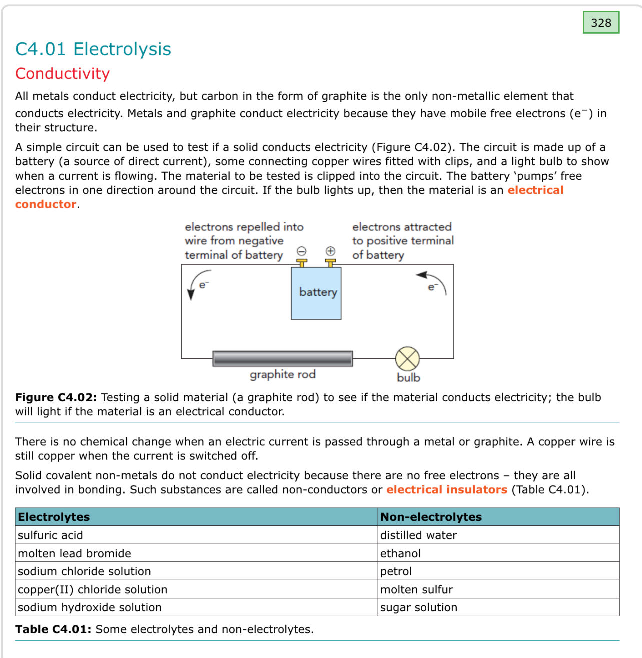
Electrochemistry is a high-value exam topic because it combines knowledge of ionic bonding, chemical equations, and practical observations. You must be able to predict products, write half-equations, and explain charge transfer — these are common exam questions worth significant marks.

2. Key Concepts

2.1 Electrical Conductivity

- **Metals** conduct electricity because they have **mobile free electrons** (e^-) in their structure. Carbon in the form of **graphite** also conducts (it is the only non-metallic element that does).
- When a current passes through a metal or graphite, **no chemical change** occurs. A copper wire remains copper.
- **Covalent substances** (e.g., plastic, sugar, ethanol) do NOT conduct electricity — they have no free electrons. These are called **electrical insulators** or **non-conductors**.

Key distinction for exams: Conductivity in metals = electron flow, NO chemical change. Conductivity in electrolytes = ion flow, chemical change DOES occur.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C4.01 Electrolysis

Conductivity

All metals conduct electricity, but carbon in the form of graphite is the only non-metallic element that conducts electricity. Metals and graphite conduct electricity because they have mobile free electrons (e^-) in their structure.

A simple circuit can be used to test if a solid conducts electricity (Figure C4.02). The circuit is made up of a battery (a source of direct current), some connecting copper wires fitted with clips, and a light bulb to show when a current is flowing. The material to be tested is clipped into the circuit. The battery 'pumps' free electrons in one direction around the circuit. If the bulb lights up, then the material is an **electrical conductor**.

Figure C4.02: Testing a solid material (a graphite rod) to see if the material conducts electricity; the bulb will light if the material is an electrical conductor.

There is no chemical change when an electric current is passed through a metal or graphite. A copper wire is still copper when the current is switched off.

Solid covalent non-metals do not conduct electricity because there are no free electrons – they are all involved in bonding. Such substances are called non-conductors or **electrical insulators** (Table C4.01).

Electrolytes	Non-electrolytes
sulfuric acid	distilled water
molten lead bromide	ethanol
sodium chloride solution	petrol
copper(II) chloride solution	molten sulfur
sodium hydroxide solution	sugar solution

Table C4.01: Some electrolytes and non-electrolytes.

Figure C4.02: Testing a solid material with a simple circuit — the bulb lights if the material is an electrical conductor.

2.2 Electrolytes and Non-Electrolytes

- **Electrolytes**: liquids that conduct electricity through the movement of ions AND undergo chemical decomposition. Examples: sulfuric acid, molten lead bromide, sodium chloride solution, copper(II) chloride solution, sodium hydroxide solution.
- **Non-electrolytes**: liquids that do NOT conduct electricity. Examples: distilled water, ethanol, petrol, molten sulfur, sugar solution.

Why ionic compounds must be liquid or dissolved: In the solid state, ions are fixed in a lattice and cannot move. They must be **molten** or **dissolved in water** so the ions are free to move and carry the charge.

2.3 Electrolysis — The Core Definition

> **Electrolysis** is the decomposition of an ionic compound by the passage of an electric current when the compound is molten or dissolved in aqueous solution.

The word itself tells you: *electro-* = uses electricity; *-lysis* = to split/break down.

Example: Lead(II) bromide → lead + bromine

$PbBr_2(l) \rightarrow Pb(l) + Br_2(g)$

2.4 The Electrolytic Cell — Components

An **electrolytic cell** consists of:

- **Power supply** (battery or power pack) — provides direct current (DC)
- **Electrolyte** — the molten or dissolved ionic compound
- **Two electrodes** dipped into the electrolyte:
- **Cathode (−)** — the negative electrode, connected to the negative terminal of the battery
- **Anode (+)** — the positive electrode, connected to the positive terminal of the battery
- **Inert electrodes** — usually graphite or platinum; chosen because they do not react with the electrolyte or products

Memory aid: **C**athode = **C**ations go here = **N**egative electrode. Think "CaN" — Cathode Attracts Negative-charge-seeking cations (positive ions go to the negative electrode).

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Some liquids conduct electricity. The process in which an electric current flows through a liquid compound or solution is called **electrolysis**. Unlike electrical conductivity in metals, electrolysis in liquid compounds or solutions produces a chemical reaction, causing the substance to change.

The conductivity of liquids can be tested in a similar way to solids, but the simple testing circuit is changed (Figure C4.03). Instead of clipping the solid material to be tested into the circuit, graphite rods are dipped into the test liquid. Liquid compounds, solutions and molten materials can all be tested in this way.

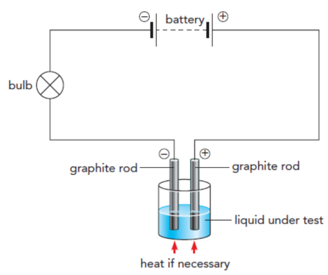

Figure C4.03: Apparatus for testing the conductivity of liquids (an electrolytic cell).

If liquid compounds or solutions are tested using the apparatus in Figure C4.03, then the result will depend on the type of bonding in the compound (Chapter C2). Molten metals, and mercury, which is liquid at room temperature, conduct electricity. Electrons are still able to move through the liquid metal to carry the charge. As in solid metals, no chemical change takes place when liquid metals conduct electricity. If the compound is bonded covalently, then the compound will *not* conduct electricity as a liquid or as a solution.

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Examples of such liquids are ethanol, petrol (gasoline), pure water and sugar solution. Ionic compounds *will* conduct electricity if they are either molten or dissolved in water. Examples of these ionic compounds are molten lead bromide, sodium chloride solution and copper(II) sulfate solution.

Figure C4.03: Apparatus for testing the conductivity of liquids — an electrolytic cell with graphite rod electrodes.

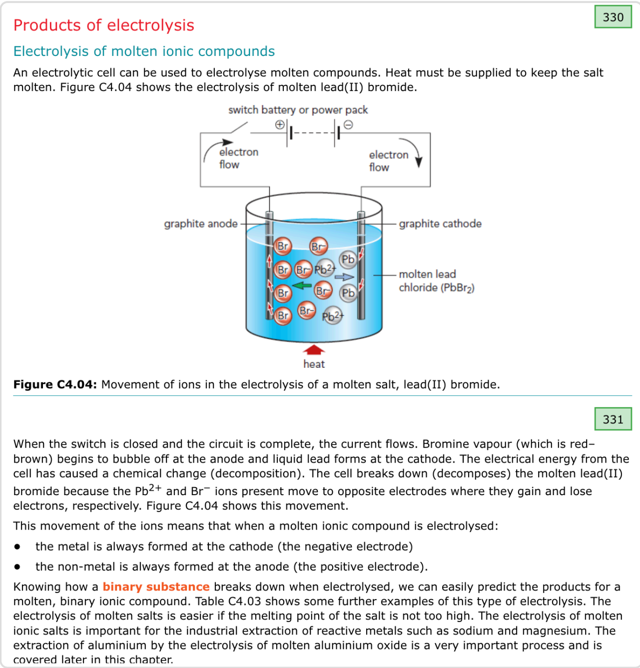
2.5 Movement of Ions During Electrolysis

During electrolysis:

- **Cations** (positive ions — metal ions or H^+) move towards the **cathode (−)**
- **Anions** (negative ions — non-metal ions) move towards the **anode (+)**

Charge transfer happens in four simultaneous ways:

1. In the **external circuit** (wires): electrons flow from the negative terminal of the battery to the cathode, then from the anode back to the positive terminal
2. At the **cathode**: electrons transfer from the electrode to cations, forming uncharged atoms (reduction)
3. At the **anode**: electrons transfer from anions to the electrode, forming uncharged atoms (oxidation)
4. In the **electrolyte**: positive ions move towards the cathode; negative ions move towards the anode



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Products of electrolysis

Electrolysis of molten ionic compounds

An electrolytic cell can be used to electrolyse molten compounds. Heat must be supplied to keep the salt molten. Figure C4.04 shows the electrolysis of molten lead(II) bromide.

Figure C4.04: Movement of ions in the electrolysis of a molten salt, lead(II) bromide.

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When the switch is closed and the circuit is complete, the current flows. Bromine vapour (which is red–brown) begins to bubble off at the anode and liquid lead forms at the cathode. The electrical energy from the cell has caused a chemical change (decomposition). The cell breaks down (decomposes) the molten lead(II) bromide because the Pb^{2+} and Br^- ions present move to opposite electrodes where they gain and lose electrons, respectively. Figure C4.04 shows this movement.

This movement of the ions means that when a molten ionic compound is electrolysed:

- the metal is always formed at the cathode (the negative electrode)
- the non-metal is always formed at the anode (the positive electrode).

Knowing how a **binary substance** breaks down when electrolysed, we can easily predict the products for a molten, binary ionic compound. Table C4.03 shows some further examples of this type of electrolysis. The electrolysis of molten salts is easier if the melting point of the salt is not too high. The electrolysis of molten ionic salts is important for the industrial extraction of reactive metals such as sodium and magnesium. The extraction of aluminium by the electrolysis of molten aluminium oxide is a very important process and is covered later in this chapter.

Figure C4.04: Movement of ions in the electrolysis of molten lead(II) bromide — Br^- ions move to the anode, Pb^{2+} ions move to the cathode.

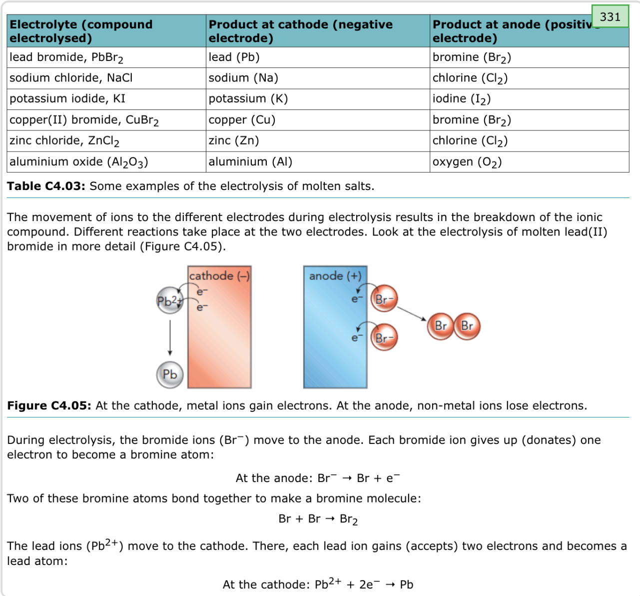
2.6 Electrolysis of Molten Ionic Compounds

When a **molten binary ionic compound** is electrolysed:

- The **metal** is always formed at the **cathode** (negative electrode)
- The **non-metal** is always formed at the **anode** (positive electrode)

Key examples (Table C4.03):

Electrolyte	Product at cathode (−)	Product at anode (+)
Lead bromide, $PbBr_2$	Lead (Pb)	Bromine (Br_2)
Sodium chloride, NaCl	Sodium (Na)	Chlorine (Cl_2)
Potassium iodide, KI	Potassium (K)	Iodine (I_2)
Copper(II) bromide, $CuBr_2$	Copper (Cu)	Bromine (Br_2)
Zinc chloride, $ZnCl_2$	Zinc (Zn)	Chlorine (Cl_2)
Aluminium oxide, Al_2O_3	Aluminium (Al)	Oxygen (O_2)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Electrolyte (compound electrolysed)	Product at cathode (negative electrode)	Product at anode (positive electrode)
lead bromide, $PbBr_2$	lead (Pb)	bromine (Br_2)
sodium chloride, NaCl	sodium (Na)	chlorine (Cl_2)
potassium iodide, KI	potassium (K)	iodine (I_2)
copper(II) bromide, $CuBr_2$	copper (Cu)	bromine (Br_2)
zinc chloride, $ZnCl_2$	zinc (Zn)	chlorine (Cl_2)
aluminium oxide (Al_2O_3)	aluminium (Al)	oxygen (O_2)

Table C4.03: Some examples of the electrolysis of molten salts.

The movement of ions to the different electrodes during electrolysis results in the breakdown of the ionic compound. Different reactions take place at the two electrodes. Look at the electrolysis of molten lead(II) bromide in more detail (Figure C4.05).

Figure C4.05: At the cathode, metal ions gain electrons. At the anode, non-metal ions lose electrons.

During electrolysis, the bromide ions (Br^-) move to the anode. Each bromide ion gives up (donates) one electron to become a bromine atom:

At the anode: $Br^- \rightarrow Br + e^-$

Two of these bromine atoms bond together to make a bromine molecule:

$Br + Br \rightarrow Br_2$

The lead ions (Pb^{2+}) move to the cathode. There, each lead ion gains (accepts) two electrons and becomes a lead atom:

At the cathode: $Pb^{2+} + 2e^- \rightarrow Pb$

Figure C4.05: At the cathode, metal ions gain electrons (reduction). At the anode, non-metal ions lose electrons (oxidation).

2.7 Half-Equations at the Electrodes

Half-equations show what happens at each electrode separately. You need to be able to construct cathode half-equations.

At the cathode (reduction — ions GAIN electrons):

Electrolyte	Cathode half-equation
Lead bromide, $PbBr_2$	$Pb^{2+} + 2e^- \rightarrow Pb$
Sodium chloride, NaCl	$Na^+ + e^- \rightarrow Na$
Aluminium oxide, Al_2O_3	$Al^{3+} + 3e^- \rightarrow Al$
Copper(II) bromide, $CuBr_2$	$Cu^{2+} + 2e^- \rightarrow Cu$

At the anode (oxidation — ions LOSE electrons):

Electrolyte	Anode half-equation
Lead bromide, $PbBr_2$	$2Br^- \rightarrow Br_2 + 2e^-$
Sodium chloride, NaCl	$2Cl^- \rightarrow Cl_2 + 2e^-$
Aluminium oxide, Al_2O_3	$2O^{2-} \rightarrow O_2 + 4e^-$

How to write a cathode half-equation:

1. Write the ion on the left
2. Add electrons (e^-) to the left to balance the charge
3. Write the metal atom on the right
4. The number of electrons = the charge on the ion

2.8 Electrolysis of Solutions (Aqueous Electrolysis)

When ionic compounds are **dissolved in water**, the products may be **different** from the molten compound. This is because water itself dissociates slightly:

$H_2O \rightarrow H^+ + OH^-$

So the solution contains **four types of ions** (two from the salt + H^+ and OH^- from water). At each electrode, only **one type of ion** is discharged.

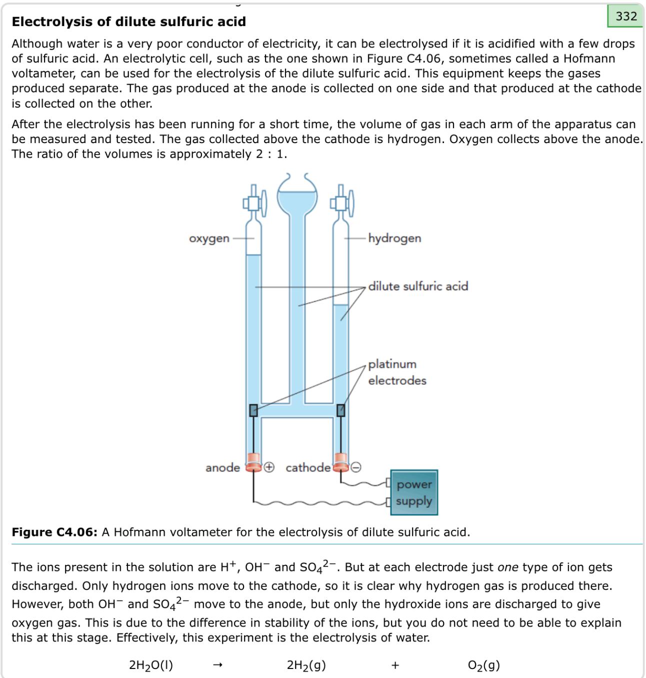
Rules for predicting products in solution:

- At the **cathode**: if the metal is MORE reactive than hydrogen, then **hydrogen** is produced (not the metal). If the metal is LESS reactive than hydrogen (e.g., copper), the **metal** is deposited.
- At the **anode**: if a halide ion (Cl^-, Br^-, I^-) is present in concentrated solution, the **halogen** is produced. Otherwise, **oxygen** is produced from hydroxide ions.

2.9 Electrolysis of Dilute Sulfuric Acid

- Ions present: H^+, OH^-, SO_4^{2-}
- At the **cathode**: hydrogen gas (H_2) — colourless gas
- At the **anode**: oxygen gas (O_2) — colourless gas
- Volume ratio: hydrogen : oxygen = **2 : 1**
- This is effectively the electrolysis of water:

$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$



Electrolysis of dilute sulfuric acid 332

Although water is a very poor conductor of electricity, it can be electrolysed if it is acidified with a few drops of sulfuric acid. An electrolytic cell, such as the one shown in Figure C4.06, sometimes called a Hofmann voltameter, can be used for the electrolysis of the dilute sulfuric acid. This equipment keeps the gases produced separate. The gas produced at the anode is collected on one side and that produced at the cathode is collected on the other.

After the electrolysis has been running for a short time, the volume of gas in each arm of the apparatus can be measured and tested. The gas collected above the cathode is hydrogen. Oxygen collects above the anode. The ratio of the volumes is approximately 2 : 1.

Figure C4.06: A Hofmann voltameter for the electrolysis of dilute sulfuric acid.

The ions present in the solution are H^+, OH^- and SO_4^{2-}. But at each electrode just *one* type of ion gets discharged. Only hydrogen ions move to the cathode, so it is clear why hydrogen gas is produced there. However, both OH^- and SO_4^{2-} move to the anode, but only the hydroxide ions are discharged to give oxygen gas. This is due to the difference in stability of the ions, but you do not need to be able to explain this at this stage. Effectively, this experiment is the electrolysis of water.

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

Figure C4.06: A Hofmann voltameter for the electrolysis of dilute sulfuric acid — hydrogen collects at the cathode, oxygen at the anode, in a 2:1 volume ratio.

2.10 Electrolysis of Concentrated Sodium Chloride Solution

- Ions present: Na^+, H^+, Cl^-, OH^-
- At the **cathode**: **hydrogen** gas (H_2) — not sodium, because sodium is more reactive than hydrogen
- At the **anode**: **chlorine** gas (Cl_2) — a green gas with a pungent smell

Comparison (Table C4.05):

Condition	Product at cathode	Product at anode
Molten NaCl	Sodium (Na)	Chlorine (Cl_2)
Concentrated NaCl(aq)	Hydrogen (H_2)	Chlorine (Cl_2)

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Electrolysis of sodium chloride solution

A concentrated solution of sodium chloride can be electrolysed in the laboratory (Figure C4.07). There are four different ions present in the solution. The positive ions (cations), Na^+ and H^+, flow to the cathode, attracted by its negative charge. The negative ions (anions), Cl^- and OH^-, travel to the anode.

When this experiment is carried out, hydrogen is produced at the cathode, not sodium. Chlorine, a green gas, is collected at the anode. This chlorine is produced by the discharge of chloride ions at the anode. In this case, the hydroxide ions are not discharged and so no oxygen gas is produced.

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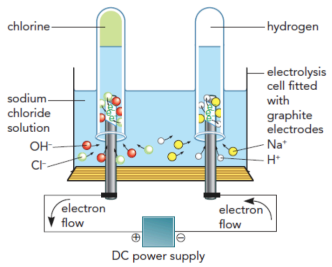

Figure C4.07: Movement and discharge of ions in the electrolysis of concentrated sodium chloride solution.

It is important to understand that different products are formed when concentrated sodium chloride solution is electrolysed rather than molten sodium chloride (Table C4.05).

Electrolyte	Product at cathode	Product at anode
molten sodium chloride, NaCl(l)	sodium (Na)	chlorine (Cl_2)
concentrated sodium chloride solution, NaCl(aq)	hydrogen (H_2)	chlorine (Cl_2)

Table C4.05: Products of electrolysis of sodium chloride under different conditions.

Figure C4.07: Movement and discharge of ions in the electrolysis of concentrated sodium chloride solution — chlorine at the anode, hydrogen at the cathode.

2.11 Electrolysis of Copper(II) Sulfate Solution

With inert (graphite) electrodes:

- Ions present: Cu^{2+}, H^+, SO_4^{2-}, OH^-
- At the cathode: **copper** metal deposited (red-brown coating) — because copper is less reactive than hydrogen
- $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
- At the anode: **oxygen** gas produced — hydroxide ions are discharged rather than sulfate ions
- $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$
- **Observations**: blue colour of solution **fades** over time (Cu^{2+} ions removed); solution becomes more **acidic** (OH^- ions removed)

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Electrolysis of copper(II) sulfate

Further information on the nature of electrode reactions can be found from experiments using different salt solutions. The electrolysis of a blue copper(II) sulfate solution using inert graphite electrodes (Figure C4.08) produces a deposit of red–brown copper metal on the cathode. Copper is less reactive than hydrogen and therefore it is the copper ions (Cu^{2+}) that are discharged at the cathode.

At the cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Oxygen gas is produced at the anode as the hydroxide ions from water are discharged rather than the sulfate ions.

At the anode: $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

As the electrolysis proceeds, the blue colour of the solution will fade as the copper ions causing the colour are discharged. The electrolyte solution will also become more acidic as OH^- ions are discharged.

Figure C4.08: Copper is quite unreactive so it can be seen deposited on the cathode when copper(II) sulfate solution is electrolysed.

In comparison to this reaction, it is possible to use electrodes other than graphite or platinum; electrodes that are not inert. A cell can be set up to electrolyse copper(II) sulfate solution using copper electrodes (not graphite electrodes). As electrolysis takes place, the cathode gains mass as copper is deposited on the electrode (Figure C4.09).

At the cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

The anode, however, loses mass as copper dissolves from the electrode:

At the anode: $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

Figure C4.08: Copper deposited on the cathode during electrolysis of copper(II) sulfate solution with graphite electrodes.

With copper electrodes:

- At the cathode: copper is **deposited** (cathode gains mass)
- $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

- At the anode: copper **dissolves** from the electrode (anode loses mass)
- $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$
- The Cu^{2+} concentration stays the same, so the **blue colour does NOT change**
- This is the basis of **electroplating** and **purification of copper**

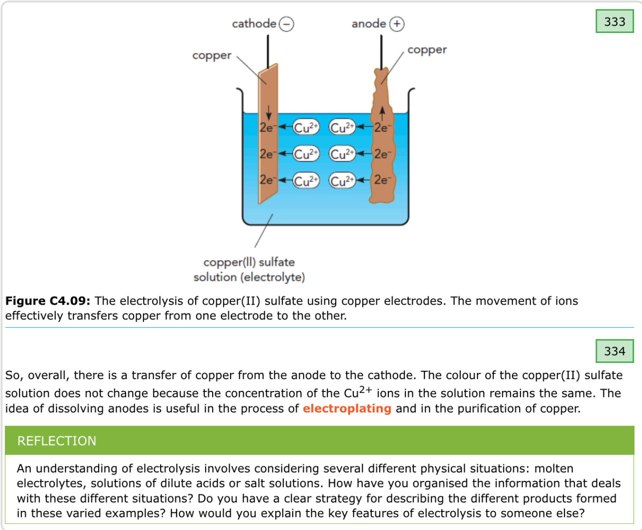

Figure C4.09: The electrolysis of copper(II) sulfate using copper electrodes. The movement of ions effectively transfers copper from one electrode to the other.

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So, overall, there is a transfer of copper from the anode to the cathode. The colour of the copper(II) sulfate solution does not change because the concentration of the Cu^{2+} ions in the solution remains the same. The idea of dissolving anodes is useful in the process of electroplating and in the purification of copper.

REFLECTION

An understanding of electrolysis involves considering several different physical situations: molten electrolytes, solutions of dilute acids or salt solutions. How have you organised the information that deals with these different situations? Do you have a clear strategy for describing the different products formed in these varied examples? How would you explain the key features of electrolysis to someone else?

Figure C4.09: Electrolysis of copper(II) sulfate using copper electrodes — copper transfers from anode to cathode, solution stays blue.

2.12 Hydrogen–Oxygen Fuel Cells

A **fuel cell** converts chemical energy directly into electrical energy. It operates continuously as long as reactants (fuel) are supplied — unlike batteries, which run down.

The reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

- Hydrogen enters at the **negative electrode**
- Oxygen enters at the **positive electrode**
- **Water** is the only product
- The cell uses an electrolyte and carbon electrodes

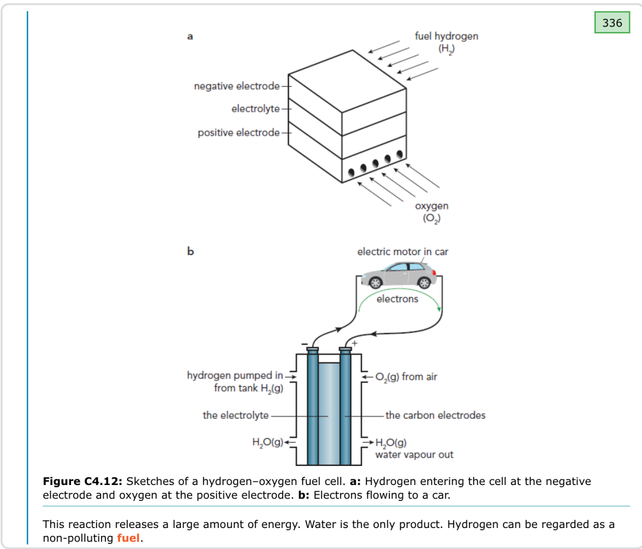
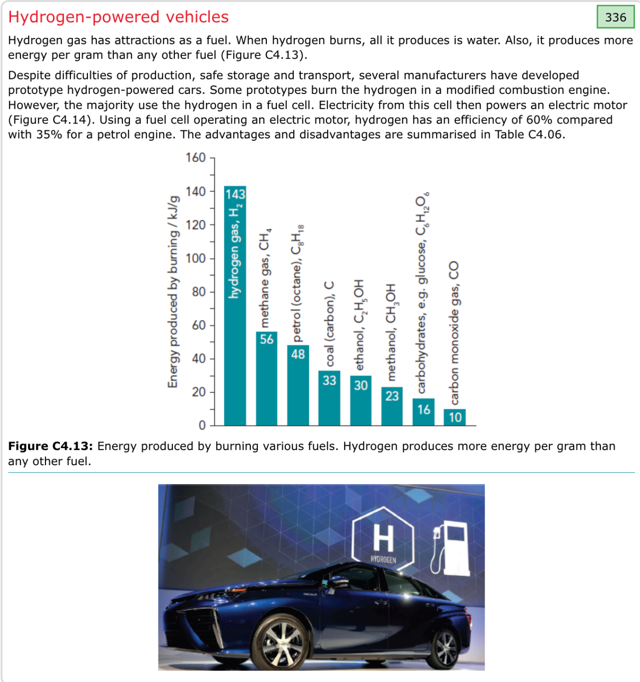


Figure C4.12: Sketches of a hydrogen–oxygen fuel cell. **a:** Hydrogen entering the cell at the negative electrode and oxygen at the positive electrode. **b:** Electrons flowing to a car.

This reaction releases a large amount of energy. Water is the only product. Hydrogen can be regarded as a non-polluting fuel.

Figure C4.12: Sketches of a hydrogen–oxygen fuel cell. (a) Hydrogen at the negative electrode, oxygen at the positive. (b) Electrons flowing to power a car.

Efficiency: A fuel cell powering an electric motor achieves **~60% efficiency**, compared to **~35% for a petrol engine**.



Hydrogen-powered vehicles 336

Hydrogen gas has attractions as a fuel. When hydrogen burns, all it produces is water. Also, it produces more energy per gram than any other fuel (Figure C4.13).

Despite difficulties of production, safe storage and transport, several manufacturers have developed prototype hydrogen-powered cars. Some prototypes burn the hydrogen in a modified combustion engine. However, the majority use the hydrogen in a fuel cell. Electricity from this cell then powers an electric motor (Figure C4.14). Using a fuel cell operating an electric motor, hydrogen has an efficiency of 60% compared with 35% for a petrol engine. The advantages and disadvantages are summarised in Table C4.06.

Figure C4.13: Energy produced by burning various fuels. Hydrogen produces more energy per gram than any other fuel.

Figure C4.13: Energy produced by burning various fuels — hydrogen produces 143 kJ/g, more than any other fuel.

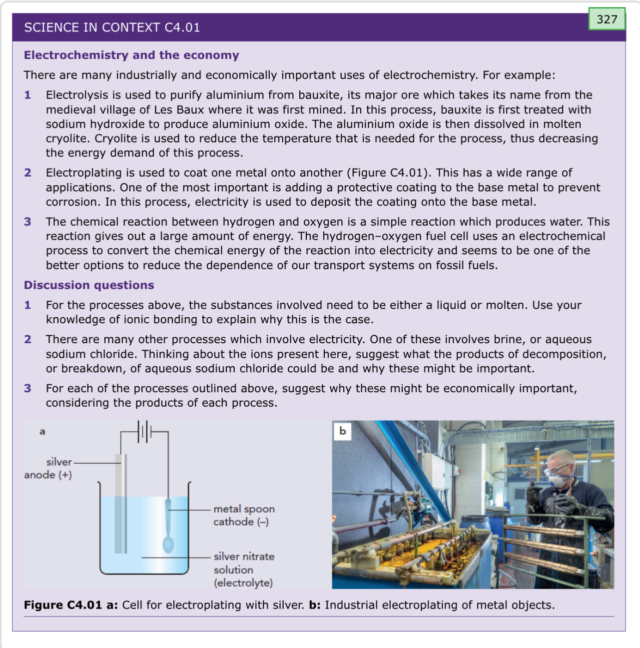
2.13 Advantages and Disadvantages of Hydrogen Fuel Cells

Advantages	Disadvantages
Renewable if produced using solar/wind energy	Non-renewable if generated using nuclear/fossil fuels
Lower flammability than petrol	Large fuel tank required
Virtually emission-free	Very few hydrogen filling stations
Zero CO_2 emissions	Engine redesign or fuel cell system needed
Non-toxic	Currently expensive

3. Important Terms and Definitions

Term	Definition
Electrical conductor	A material that allows electric current to flow through it (metals, graphite)
Electrical insulator	A material that does not allow electric current to flow (plastics, ceramics)
Electrolysis	Decomposition of an ionic compound by passing an electric current through it when molten or in solution
Electrolyte	A liquid that conducts electricity due to the movement of ions and is decomposed
Non-electrolyte	A liquid that does not conduct electricity
Electrolytic cell	The apparatus used to carry out electrolysis
Cathode	The negative electrode (attracts cations)
Anode	The positive electrode (attracts anions)

Cation	A positive ion (metal ion or H^+) — moves to the cathode
Anion	A negative ion (non-metal ion) — moves to the anode
Inert electrode	An electrode that does not react with the electrolyte or products (e.g., graphite, platinum)
Half-equation	An equation showing the reaction at one electrode only
Binary substance	A compound made of only two elements
Decomposition	Breaking down a compound into simpler substances
Fuel cell	A device that converts chemical energy of a fuel directly into electrical energy
Electroplating	Coating a metal object with a thin layer of another metal using electrolysis


SCIENCE IN CONTEXT C4.01

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Electrochemistry and the economy

There are many industrially and economically important uses of electrochemistry. For example:

1 Electrolysis is used to purify aluminium from bauxite, its major ore which takes its name from the medieval village of Les Baux where it was first mined. In this process, bauxite is first treated with sodium hydroxide to produce aluminium oxide. The aluminium oxide is then dissolved in molten cryolite. Cryolite is used to reduce the temperature that is needed for the process, thus decreasing the energy demand of this process.

2 Electroplating is used to coat one metal onto another (Figure C4.01). This has a wide range of applications. One of the most important is adding a protective coating to the base metal to prevent corrosion. In this process, electricity is used to deposit the coating onto the base metal.

3 The chemical reaction between hydrogen and oxygen is a simple reaction which produces water. This reaction gives out a large amount of energy. The hydrogen–oxygen fuel cell uses an electrochemical process to convert the chemical energy of the reaction into electricity and seems to be one of the better options to reduce the dependence of our transport systems on fossil fuels.

Discussion questions

1 For the processes above, the substances involved need to be either a liquid or molten. Use your knowledge of ionic bonding to explain why this is the case.

2 There are many other processes which involve electricity. One of these involves brine, or aqueous sodium chloride. Thinking about the ions present here, suggest what the products of decomposition, or breakdown, of aqueous sodium chloride could be and why these might be important.

3 For each of the processes outlined above, suggest why these might be economically important, considering the products of each process.

Figure C4.01 a: Cell for electroplating with silver. **b:** Industrial electroplating of metal objects.

Figure C4.01: (a) Cell for electroplating with silver — silver anode, metal spoon cathode, silver nitrate solution. (b) Industrial electroplating.

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4. Key Equations and Formulas

Electrolysis of molten lead(II) bromide:

- **Overall:** $PbBr_2(l) \rightarrow Pb(l) + Br_2(g)$
- **Cathode:** $Pb^{2+} + 2e^- \rightarrow Pb$
- **Anode:** $2Br^- \rightarrow Br_2 + 2e^-$

Electrolysis of dilute sulfuric acid (water):

- **Overall:** $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$
- **Cathode:** $4H^+ + 4e^- \rightarrow 2H_2$
- **Anode:** $2O^{2-} \rightarrow O_2 + 4e^-$

Electrolysis of copper(II) sulfate with graphite electrodes:

- **Cathode:** $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
- **Anode:** $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

Electrolysis of copper(II) sulfate with copper electrodes:

- **Cathode:** $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
- **Anode:** $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

Hydrogen fuel cell:

- **Overall:** $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

Water dissociation:

- $H_2O \rightarrow H^+ + OH^-$

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5. Exam Focus Points (Ranked by Likelihood)

1. **Writing cathode half-equations** — Practise for Pb^{2+}, Cu^{2+}, Na^+, Al^{3+}, H^+. Remember: number of electrons = charge on the ion.
2. **Predicting products of electrolysis** — Know the rules for molten vs. aqueous. Metal at cathode for molten; hydrogen at cathode for reactive metals in solution.
3. **Explaining why ionic compounds must be molten or dissolved** — Ions must be free to move. In solids, ions are fixed in a lattice.
4. **Copper sulfate electrolysis: graphite vs. copper electrodes** — This is a classic comparison question. Know the observations and explain why the solution fades/doesn't fade.
5. **Describing charge transfer** — Four stages: electron flow in wires, electron transfer at cathode, electron transfer at anode, ion movement in electrolyte.
6. **Advantages and disadvantages of hydrogen fuel cells** — Often a 2–3 mark written answer.
7. **Distinguishing conductors, insulators, electrolytes, and non-electrolytes** — Know examples of each.
8. **The electrolysis of sodium chloride: molten vs. solution** — Different products at the cathode.
9. **Electroplating applications** — Know the setup: object to be plated = cathode, plating metal = anode, solution of plating metal salt = electrolyte.

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6. Practice Questions with Full Answers

Question C4.01

a) Which of the following will conduct electricity?

- i. a strip of copper metal
- ii. a solution of sugar in water
- iii. aqueous sodium chloride solution
- iv. mercury
- v. dilute hydrochloric acid
- vi. melted sugar

Answer: i (copper — metal with free electrons), iii (sodium chloride solution — dissolved ionic compound with free ions), iv (mercury — liquid metal with free electrons), v (dilute hydrochloric acid — dissolved acid with free H^+ and Cl^- ions).

b) Which of i–vi are electrolytes?

Answer: iii (aqueous sodium chloride) and v (dilute hydrochloric acid). These are the ones that conduct AND undergo chemical decomposition. Mercury and copper conduct but are NOT electrolytes because no chemical change occurs.

Question C4.02

a) Why is solid potassium bromide a non-conductor of electricity?

Answer: In solid potassium bromide, the K^+ and Br^- ions are held in fixed positions in a lattice structure. They cannot move, so they cannot carry an electric charge through the substance.

b) Suggest two things that could make potassium bromide conduct electricity.

Answer:

1. **Melt it** — heating to its melting point breaks the lattice and allows the ions to move freely.
2. **Dissolve it in water** — the ions separate and become free to move in solution.

Question C4.03

Explain the major difference between electrical conductivity in metals and solutions.

Answer: In metals, the current is carried by the flow of **free electrons** and no chemical change occurs. In solutions (electrolytes), the current is carried by the movement of **ions** (cations to the cathode, anions to the anode) and a chemical change (decomposition) takes place.

Question C4.04

What products will you get when passing an electric current through:

a) A concentrated solution of sodium chloride?

Answer: Cathode: **hydrogen** gas. Anode: **chlorine** gas. (Sodium is more reactive than hydrogen, so hydrogen is produced at the cathode instead of sodium.)

b) Molten potassium chloride?

Answer: Cathode: **potassium** metal. Anode: **chlorine** gas. (When molten, there is no water present, so the metal itself is formed.)

Question C4.05

Give a general rule for predicting the electrode products when a molten binary ionic compound is electrolysed.

Answer: The metal is always produced at the cathode (negative electrode) and the non-metal is always produced at the anode (positive electrode).

Question C4.06

In the electrolysis of molten lead(II) bromide, the cathode reaction is: $Pb^{2+} + 2e^- \rightarrow Pb$

a) Write the equation for the reaction at the positive electrode.

Answer: $2Br^- \rightarrow Br_2 + 2e^-$

b) What observations would be made during this reaction?

Answer:

- At the cathode: silvery liquid **lead** forms at the bottom of the container
- At the anode: **red-brown bromine vapour** bubbles off
- The electrolyte must be heated to keep the lead bromide molten

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Question C4.07

Balance the half-equation: ___ $H_2(g) \rightarrow$ ___ $H^+(aq)$ + ___ e^- so it adds up with $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ to give the overall equation $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$.

Answer: $2H_2(g) \rightarrow 4H^+(aq) + 4e^-$

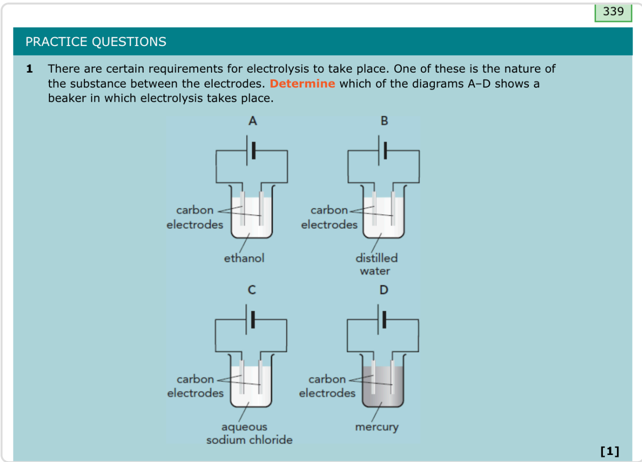
Working: The second equation uses $4H^+$ and $4e^-$, so the first equation must produce $4H^+$ and $4e^-$. Since each H_2 molecule gives $2H^+ + 2e^-$, we need $2H_2$ molecules.

Question C4.08

Sort statements a–f into fuel cell advantages and petrol car advantages:

Fuel cell car advantages	Petrol car advantages
a. Virtually emission free	b. Smaller fuel tank needed
d. Non-toxic	c. Plentiful filling stations
e. Can be renewable	f. Less expensive

Practice Question 1

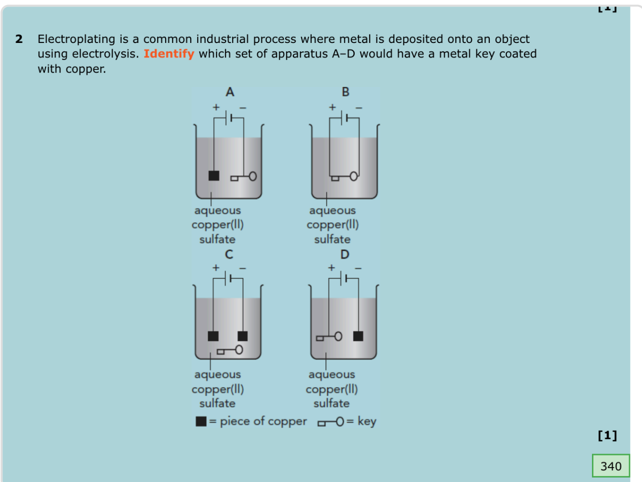


Practice Question 1: Diagrams A–D showing four beakers with carbon electrodes and different substances.

Question: There are certain requirements for electrolysis to take place. One of these is the nature of the substance between the electrodes. Determine which of the diagrams A–D shows a beaker in which electrolysis takes place.

Answer: C — aqueous sodium chloride. Electrolysis requires an ionic compound dissolved in water or molten. Ethanol is covalent (no ions). Distilled water has too few ions. Mercury is a metal (conducts but no decomposition). Only sodium chloride solution has free ions that will be decomposed.

Practice Question 2

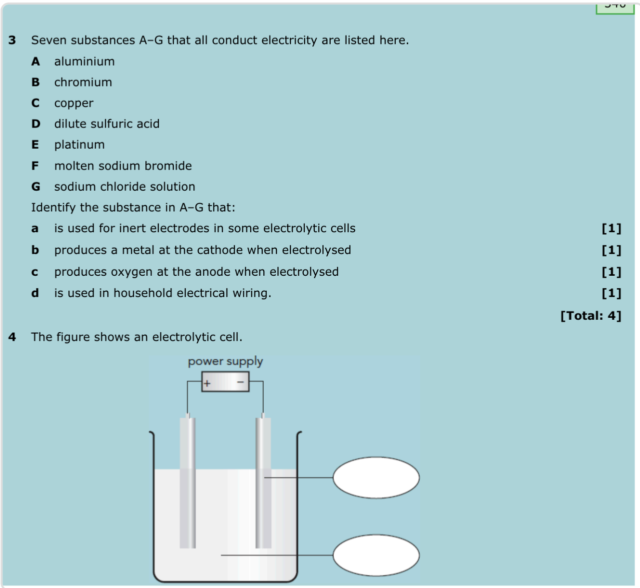


Practice Question 2: Diagrams A–D showing four electroplating setups with a piece of copper and a metal key in copper(II) sulfate solution.

Question: Electroplating is a common industrial process where metal is deposited onto an object using electrolysis. Identify which set of apparatus A–D would have a metal key coated with copper.

Answer: D — The key must be the **cathode** (negative electrode) and a piece of copper must be the **anode** (positive electrode), in a solution of copper(II) sulfate. In D, the key is connected to the negative terminal and the copper piece is connected to the positive terminal.

Practice Questions 3 and 4



3 Seven substances A–G that all conduct electricity are listed here.

A aluminium
B chromium
C copper
D dilute sulfuric acid
E platinum
F molten sodium bromide
G sodium chloride solution

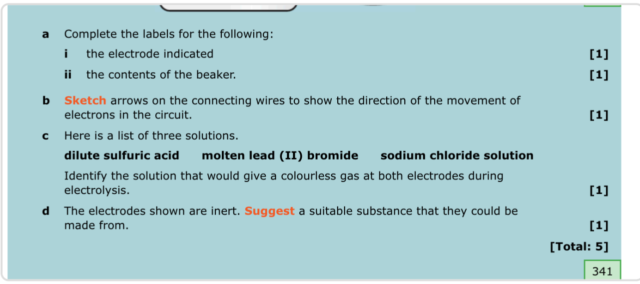
Identify the substance in A–G that:

a is used for inert electrodes in some electrolytic cells **[1]**
b produces a metal at the cathode when electrolysed **[1]**
c produces oxygen at the anode when electrolysed **[1]**
d is used in household electrical wiring. **[1]**

[Total: 4]

4 The figure shows an electrolytic cell.

Practice Questions 3 and 4: Question 3 lists seven substances A–G. Question 4 shows an electrolytic cell diagram with a power supply and two electrodes.



a Complete the labels for the following:
i the electrode indicated **[1]**
ii the contents of the beaker. **[1]**

b **Sketch** arrows on the connecting wires to show the direction of the movement of electrons in the circuit. **[1]**

c Here is a list of three solutions.

dilute sulfuric acid **molten lead (II) bromide** **sodium chloride solution**

Identify the solution that would give a colourless gas at both electrodes during electrolysis. **[1]**

d The electrodes shown are inert. **Suggest** a suitable substance that they could be made from. **[1]**

[Total: 5]

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Practice Question 4 continued: Parts a–d asking about electrode labels, beaker contents, electron flow direction, and suitable inert electrode material.

Practice Question 3

Seven substances A–G conduct electricity. Identify the substance that:

a) Is used for inert electrodes: E — Platinum (unreactive, does not interfere with electrolysis)

b) Produces a metal at the cathode when electrolysed: F — Molten sodium bromide (molten ionic compound → metal at cathode)

c) Produces oxygen at the anode when electrolysed: D — Dilute sulfuric acid (electrolysis of dilute acid produces O_2 at the anode from OH^- ions)

d) Is used in household electrical wiring: C — Copper (excellent conductor, ductile, relatively cheap)

Practice Question 4

An electrolytic cell diagram. Complete labels and answer questions.

a.i) The electrode indicated: This is the **cathode** (negative electrode) or **anode** (positive electrode) — identify from the + and − terminals on the power supply.

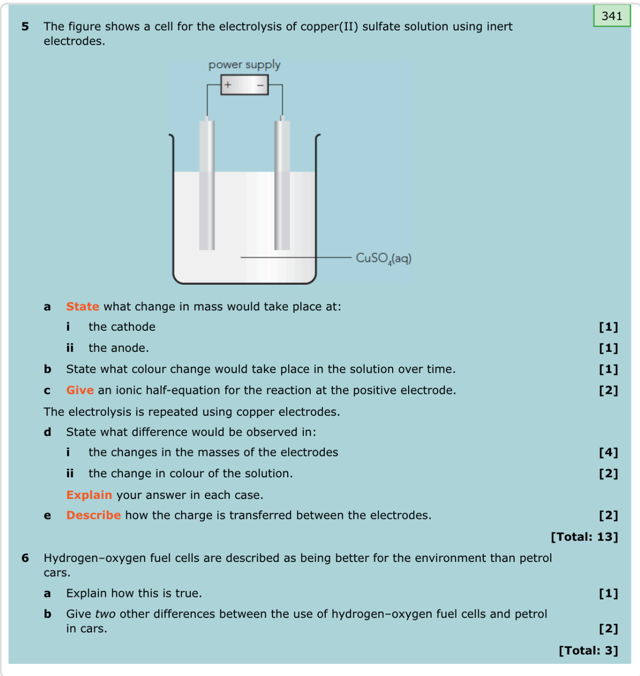
a.ii) Contents of beaker: The liquid is the **electrolyte** (e.g., a molten salt or aqueous ionic solution).

b) Direction of electron flow: Electrons flow from the **negative terminal** of the power supply through the wire to the cathode, and from the anode back through the wire to the positive terminal.

c) Which solution gives a colourless gas at BOTH electrodes: Dilute sulfuric acid — produces hydrogen (colourless) at cathode and oxygen (colourless) at anode. Molten lead bromide gives a metal and brown vapour. Sodium chloride solution gives hydrogen (colourless) but chlorine (green) at the anode.

d) Suitable substance for inert electrodes: Graphite (or platinum) — unreactive, will not react with the electrolyte or products.

Practice Questions 5 and 6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5 The figure shows a cell for the electrolysis of copper(II) sulfate solution using inert electrodes.

a State what change in mass would take place at:
- **i** the cathode [1]
- **ii** the anode. [1]

b State what colour change would take place in the solution over time. [1]

c Give an ionic half-equation for the reaction at the positive electrode. [2]

The electrolysis is repeated using copper electrodes.

d State what difference would be observed in:
- **i** the changes in the masses of the electrodes [4]
- **ii** the change in colour of the solution. [2]

Explain your answer in each case.

e Describe how the charge is transferred between the electrodes. [2]

[Total: 13]

6 Hydrogen–oxygen fuel cells are described as being better for the environment than petrol cars.

a Explain how this is true. [1]

b Give *two* other differences between the use of hydrogen–oxygen fuel cells and petrol in cars. [2]

[Total: 3]

Practice Questions 5 and 6: Question 5 shows a cell for electrolysis of $CuSO_4$(aq) with inert electrodes (13 marks). Question 6 asks about hydrogen fuel cells vs petrol cars (3 marks).

Practice Question 5 (13 marks)

Electrolysis of copper(II) sulfate with inert electrodes:

a.i) Mass change at cathode: The cathode **gains mass** — copper metal is deposited on it as Cu^{2+} ions are reduced.

a.ii) Mass change at anode: The anode **does not change mass** (it is inert). However, oxygen gas bubbles off.

b) Colour change in solution: The blue colour **fades/becomes paler** over time because Cu^{2+} ions (which are blue) are removed from solution and deposited as copper.

c) Ionic half-equation at the positive electrode: $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

d.i) With copper electrodes — mass changes: The cathode **gains mass** (copper deposited) AND the anode **loses mass** (copper dissolves). The mass gained by the cathode equals the mass lost by the anode because Cu^{2+} is deposited at one and released at the other.

d.ii) With copper electrodes — colour change: No colour change — the concentration of Cu^{2+} ions remains constant because for every Cu^{2+} deposited at the cathode, another Cu^{2+} is released from the anode.

e) How charge is transferred between electrodes:

1. In the external wires, charge is carried by **electrons** flowing from the anode to the cathode (via the battery).
2. In the electrolyte, charge is carried by **ions** — Cu^{2+} cations move towards the cathode, and SO_4^{2-} and OH^- anions move towards the anode.

3. At the cathode, electrons are transferred from the electrode to Cu^{2+} ions (reduction).
4. At the anode, electrons are transferred from OH^- ions to the electrode (oxidation).

Practice Question 6 (3 marks)

Hydrogen–oxygen fuel cells are better for the environment than petrol cars.

a) Explain how (1 mark): The only product of a hydrogen fuel cell is **water**, so there are **zero harmful emissions** (no CO_2, no CO, no particulates, no NO_x). Petrol cars produce CO_2 and other pollutants from combustion.

b) Two other differences (2 marks):

1. Fuel cells are more **energy-efficient** (~60%) compared to petrol engines (~35%).
2. Hydrogen fuel is **non-toxic** whereas petrol is toxic and carcinogenic.

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7. Quick Revision Card

Electrolysis = decomposition of ionic compound by electric current (must be molten or dissolved)

- **Cathode (−)**: attracts cations → metals or hydrogen formed
- **Anode (+)**: attracts anions → non-metals formed
- **Inert electrodes**: graphite or platinum (don't react)

Molten salts → simple prediction: metal at cathode, non-metal at anode

Solutions → tricky because of water: H^+ and OH^- also present

- Cathode: reactive metals → hydrogen; less reactive metals (Cu) → metal deposited
- Anode: halide present → halogen; no halide → oxygen

Must-know half-equations:

- $Pb^{2+} + 2e^- \rightarrow Pb$
- $Cu^{2+} + 2e^- \rightarrow Cu$
- $2Br^- \rightarrow Br_2 + 2e^-$
- $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$

$CuSO_4$ electrolysis — THE classic comparison:

- Graphite electrodes: Cu deposited, O_2 at anode, solution fades
- Copper electrodes: Cu deposited, Cu dissolves, solution stays blue

Fuel cells:

- $2H_2 + O_2 \rightarrow 2H_2O$ (only product = water)
- 60% efficient vs 35% for petrol
- Advantages: clean, non-toxic, renewable (if green hydrogen)
- Disadvantages: expensive, few filling stations, large tank needed

Charge transfer — 4 parts (common exam question):

1. Electrons in wires
2. Electron transfer at cathode (gain)
3. Electron transfer at anode (loss)
4. Ion movement in electrolyte

PROJECT C4.01 GENERATING 'GREEN' HYDROGEN FOR MODERN TRANSPORT SYSTEMS

You work for a scientific consultancy and have been asked to explain the process of electrolysis to members of the government to help inform their decisions to establish hydrogen-generating plants in several regions of your country.

Work in groups to prepare a presentation on the process of electrolysis and its environmental impacts. In your presentation, include the following:

- An explanation of the process of electrolysing water to produce hydrogen. Use the structure provided by the explanatory triangle shown in Figure C4.15. This helps to explain the meaning of the word electrolysis and the process involved. Use the example of the electrolysis of dilute sulfuric acid in your explanation.
- Identify the sources of clean energy needed to carry out the electrolysis.
- Suggest how the hydrogen is used to power different types of transport system, and their environmental advantage.

Share your presentations with the whole class.

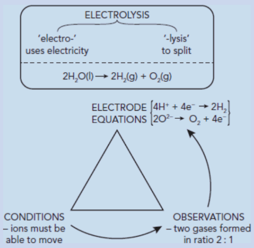

Figure C4.15: The electrolysis of water.

Figure C4.15: Summary triangle — the electrolysis of water, showing conditions, observations, electrode equations, and the meaning of the word electrolysis.